

Deep-Learning-Based Automated Morphology Analysis With Atomic Force Microscopy

Yingao Chang^{ID}, Zhiang Liu^{ID}, Yinan Wu^{ID}, *Member, IEEE*, and Yongchun Fang^{ID}, *Senior Member, IEEE*

Abstract—Morphology analysis based on atomic force microscopy (AFM) imaging contributes to understanding the characteristics of specimens more deeply. The preliminary and crucial step of implementing morphology analysis is to precisely segment the target area from the complex background. In this study, an automated AFM image segmentation strategy based on a well-designed U-shaped neural network is proposed to achieve accurate and robust segmentation for AFM images of different samples, thus realizing morphology analysis in micro-nano scale. Specifically, the centralized information interaction strategy cooperated with a two-path attention module is introduced to realize efficient cross-scale information interaction, which can fundamentally avoid the negative effects induced by spatial interpolation. Besides, the global information flows are adopted to guide the global information extracted by atrous spatial pyramid pooling to each level of the top-down pathway, which ensures that the high-level semantic information is not diluted during the top-down transmission process, thus locating the target area more precisely. Moreover, an AFM image dataset is constructed to train the network, which will be available online for free to facilitate other data-based AFM research. The segmentation results demonstrate that the proposed strategy has better performance on multiple AFM images compared with traditional Otsu method, fully convolutional network and U-Net. The application of the proposed method is carried out to exhibit the effectiveness in automated morphology analysis.

Note to Practitioners—Despite the growing demand of AFM-based morphology analysis in many fields, the automated analysis is still lacking limited by accuracy and robustness of AFM image segmentation. Since manual segmentation, sometimes tedious and time-consuming, heavily depends on the personal judgment, it is thus necessary to develop automated segmentation methods. Although traditional automated segmentation algorithms have good performance on certain types of images, they may be difficult to apply in different scenarios, especially for micro-nano images, due to the limited robustness. Therefore, this paper proposes an automated image segmentation algorithm based on an improved U-shaped neural network to achieve accurate morphology analysis for AFM images. The proposed automated morphology analysis workflow will be a practical tool to help reduce human workload and subjective errors, as well as

enhancing the accuracy and robustness of the analysis process. In addition, the constructed AFM image dataset can greatly facilitate the research on data-based AFM image analysis of other practitioners. Moreover, practitioners can benefit from our algorithm to improve the accuracy and robustness of image segmentation in other practical applications.

Index Terms—Atomic force microscopy, morphology analysis, image segmentation, deep learning, U-shaped structure.

I. INTRODUCTION

THE latest research of nanotechnology increasingly relies on image data to reveal the properties and deeper mechanisms [1], [2], [3], [4]. In general, morphology analysis is an effective method to make sense of the imaging data, which commonly includes the steps of pre-processing, segmentation and further analysis [5]. As the key in morphology analysis, image segmentation is challenging since it requires to accurately extract regions of interest from complex backgrounds [6]. Based on the segmentation results, it is much easier to carry out further quantitative analysis of shape, size, angularity, curvature and so on. Therefore, the segmentation performance directly influences the subsequent analysis and may lead to different conclusions.

As a powerful tool to explore the microworld, an atomic force microscopy (AFM) has made significant progress in imaging and analytical techniques [7], [8], [9], [10], [11], which reconstructs three-dimensional images of the sample surface with nanoscale resolution [12], [13], [14], [15], [16]. Since the interatomic force is universal, an AFM can measure both conductive and nonconductive samples in air or liquid, which thus plays an important role in many studies [17], [18], [19], [20], [21]. To analyze the morphological characteristics of samples through AFM imaging data, it is crucial to segment the specimens from the background [22], [23]. Although it is possible to conduct manual segmentation, the process is often tedious and prone to human bias [24], which may result in low efficiency and inconsistency. To avoid the issue of the manual approach and provide quantitative criteria for image segmentation, it is necessary to develop automated AFM image segmentation methods.

At present, the commonly used automated AFM image segmentation methods are the threshold-based method [6] and the watershed method [25]. The threshold-based method seeks a global height threshold to classify the pixels into specimens and background. Typically, areas higher than the threshold are categorized as specimens and the lower ones are categorized as

Manuscript received 24 August 2023; revised 14 November 2023; accepted 21 December 2023. Date of publication 29 December 2023; date of current version 16 October 2024. This article was recommended for publication by Associate Editor X. Liu and Editor L. Zhang upon evaluation of the reviewers' comments. This work was supported by the National Natural Science Foundation of China under Grant 62003172 and Grant 21933006. (Corresponding author: Yinan Wu.)

The authors are with the Tianjin Key Laboratory of Intelligent Robotics, College of Artificial Intelligence, Institute of Robotics and Automatic Information System, Nankai University, Tianjin 300350, China (e-mail: wuyn@nankai.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TASE.2023.3346887>.

Digital Object Identifier 10.1109/TASE.2023.3346887

1545-5955 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See <https://www.ieee.org/publications/rights/index.html> for more information.

background. Hence, the effectiveness is sensitive to threshold selection and is limited for samples with variable morphology height [26]. The watershed algorithm treats an AFM image as a topographical map, the object segmentation process is like collecting water into basins by inverting the morphology height. In addition to the classical watershed method, some improved watershed algorithms are also applied for AFM image segmentation [27], [28]. Despite the brevity and lower complexity, the effect of the watershed algorithm is limited since it is easy to cause over-segmentation results. Both the threshold-based method and the watershed method may have good performance for high-quality images, however, they are often powerless when processing raw AFM images since vertical drift, hysteresis, cross-coupling and other factors may cause artifacts and distortion in AFM images [29]. For this reason, it is necessary to cooperate with image pre-processing algorithms to realize image segmentation, which increases the workload and still cannot guarantee the segmentation quality.

In addition, some advanced automated methods are proposed for specific AFM segmentation scenes. For example, a new image segmentation method combining the traditional threshold method with the active contour method is proposed to realize accurate morphological characterization of nanobubbles in AFM images [30]. The threshold method is first implemented to obtain the preliminary boundaries of nanobubbles, afterwards, the active contour method utilizes the height gradient to make the preliminary boundaries converge to achieve optimized boundary location. By selecting different height thresholds during contour initialization, the robustness of the method is well demonstrated. In [31], a two-step approach integrating the spherical Hough transform and the contour expansion is developed to segment nanobubbles and nanodroplets with high precision. This method has strong robustness when processing AFM images with an uneven background. In [32], the Hessian blob algorithm is combined with a scale-space framework to achieve accurate sub-pixel particle detection without the requirement of image pre-processing. Nevertheless, the above segmentation methods may only be effective for some specific scenarios and sometimes it is unclear which method is more appropriate. Furthermore, some nano research increasingly relies on large amounts of morphology images [33], which leads to a considerable workload in choosing suitable methods for massive AFM images with different kinds of patterns and noises. Hence, a generalized AFM image segmentation approach is desired for automated morphology analysis.

Owing to the development of artificial intelligence techniques, machine learning and deep learning have demonstrated remarkable performance in AFM research [34], [35], [36], [37], [38], [39], [40], [41], [42]. Especially with the progress of convolutional neural networks, computer capabilities and computer vision theory, image segmentation based on deep learning has surpassed traditional methods in terms of accuracy and robustness [43]. Inspired by the powerful feature extraction capability of U-Net and the variants for image segmentation [44], [45], [46], a general AFM image segmentation method based on deep learning is proposed to achieve automated morphology analysis for various AFM

images with high accuracy and robustness. As shown in Fig. 1, the core of the proposed strategy is an improved deep-learning-based U-shaped neural network. The network is well trained by AFM images and the corresponding labels, which is able to output accurate segmentation results via using the raw AFM images. Further analysis of shape, area and other properties of the sample can be easily characterized based on the accurately segmented specimens. Thanks to the multi-scale information interaction ability and the global information guidance capability in [47], [48], the proposed method can segment the raw AFM images accurately without the requirement of eliminating the artifacts and distortion caused by vertical drift, hysteresis and creep, which is an advantage of the proposed method since additional image processing is required for some other segmentation methods. The effectiveness of the proposed method is validated by contrast experiments with the traditional Otsu method, fully convolutional networks (FCN) [49] and U-Net [50], using multiple AFM images with different qualities and noises. The application performance verifies that the proposed method can facilitate further morphology analysis of AFM images.

The main contributions of the paper are summarized as follows: 1) A robust AFM image segmentation algorithm is proposed based on an improved deep learning network, which is able to achieve accurate feature segmentation for raw AFM images of different samples without additional image processing. 2) An automated morphology analysis workflow is designed to achieve accurate feature representation in micro-nano scale with high efficiency and accuracy. 3) A dataset containing 500 AFM images and the corresponding labels is made for image analysis, which is further open-source to promote the advancement of data-based AFM research.

The rest of the paper is organized as follows: Section II analyzes the error sources that induce artifacts and distortion in AFM images and the influence on automated AFM image segmentation. The specific network structure of the proposed method is introduced in detail in Section III. Section IV presents the implementation process, the segmentation results and the application of the proposed method. Finally, the conclusion of the paper is drawn in Section V.

II. PROBLEM ANALYSIS

A. Artifacts and Distortion

Despite the atomic resolution and three-dimensional imaging ability of an AFM, artifacts and distortion, induced by the scanning system, the tip, the hardware feedback circuit and vibrations, still inevitably appear in the obtained sample images [51]. These errors result in the artifacts of the topographic image in the vertical direction and deformation in the horizontal direction. Especially, vertical drift (Fig. 2(a)) and illusory slope (Fig. 2(b)) are the main causes of vertical artifacts, which makes it difficult to truly characterize the morphological features of sample surface. Vertical drift is mainly induced by temperature changes, environmental noises and mechanical vibrations during the scanning process, etc, which causes random displacements between the probe and the scanning stage in the vertical direction, resulting in artificial

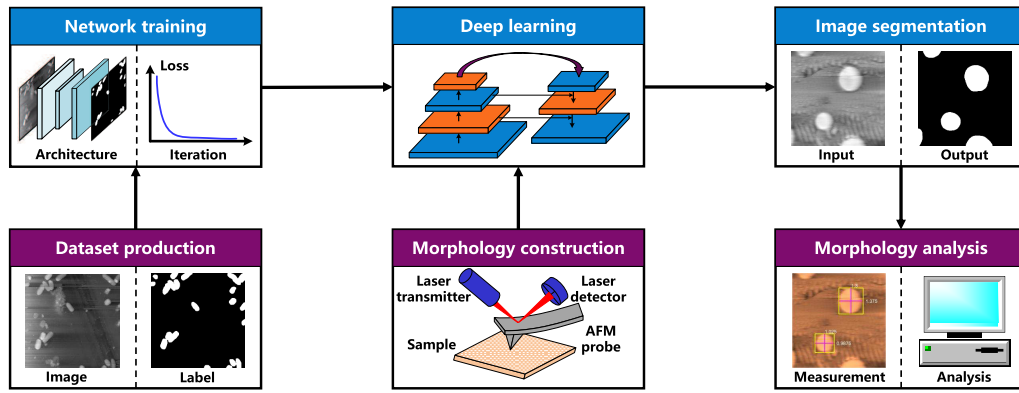


Fig. 1. The schematic diagram of the proposed automated morphology analysis workflow. The accurate segmentation result is generated by feeding the raw AFM images without pre-processing to the well-trained deep learning network, based on which morphology analysis is further carried out with high precision.

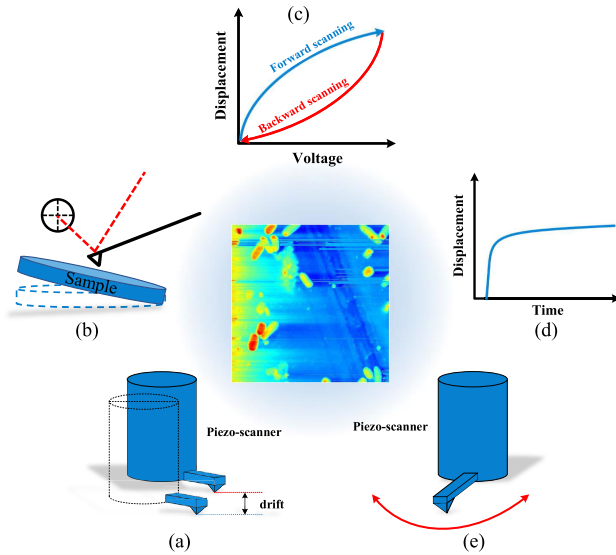


Fig. 2. Formation mechanism of the artifacts and distortion in AFM images. The causes of artifacts and distortion include (a) vertical drift, (b) illusory slope, (c) hysteresis, (d) creep and (e) intrinsic nonlinearity of the piezo-scanner.

ripples in the obtained image. The illusory slope refers to the slant angle between the horizontal scanning plane of the AFM and the sample surface, which causes an artificial tilt in the constructed topography image, making uneven brightness distribution of the final image. Furthermore, some other factors, such as hysteresis (Fig. 2(c)), creep (Fig. 2(d)), intrinsic nonlinearity of the scanner (Fig. 2(e)) and image processing also bring artifacts and distortion in AFM images. The uneven brightness, ripples and other noise make it difficult to accurately segment the specimens from the background. Sometimes, image pre-processing is required to eliminate the noise and calibrate the image to achieve better segmentation performance.

B. Lack of Automated Analysis Workflow

Although it is possible to manually segment the distorted AFM images, the workload is sometimes very large because there may be hundreds of images and each image may contain multiple specimens. On the other hand, different operators

may have different segmentation results for the same image due to different biases. While some automated analytical workflows have already been reported in other microscopy fields [52], [53], a satisfactory automated approach is still lacking in AFMs. The most widely used modular program to visualize and analyze AFM images is Gwyddion. Threshold-based method and watershed method are the two built-in segmentation methods in Gwyddion, however, the segmentation process is often conducted after performing manual image pre-processing due to the artifact and distortion of a raw AFM image.

Specifically, to segment a distorted AFM image, operators should manually conduct line correction and point correction to correct errors in the three-dimensional data and flatten the image. This may be feasible for a single image or a couple of images, but seems impractical for dealing with a large number of AFM images. Thus, there is an urgent need for a versatile, automated image segmentation solution to overcome these shortcomings.

III. IMPLEMENTATION OF THE PROPOSED METHOD

The U-shaped neural network is first proposed by Ronneberger et al. on the basis of fully convolutional neural networks in 2015 [50]. It is originally intended to solve the problem of biomedical image segmentation, and is later widely used in many scenarios of semantic segmentation due to its terrific performance and simple implementation. The network is symmetrically distributed in a U-shape structure, which is mainly made up of two parts, a bottom-up pathway and a top-down pathway, separately. In the bottom-up pathway, pooling operations are used for down-sampling to extract high-level semantic features in the target area. In the top-down pathway, through a series of interpolation and other operations, the high-level semantic information extracted from the bottom-up pathway is gradually up-sampled to the original compressed image size. Compared with the classical convolutional neural networks, the U-shaped neural network is more powerful for image segmentation tasks due to its ability to efficiently handle inputs of any size and capture both local and global information. Additionally, the advantage of utilizing fewer parameters further enhances the effectiveness of the U-shaped

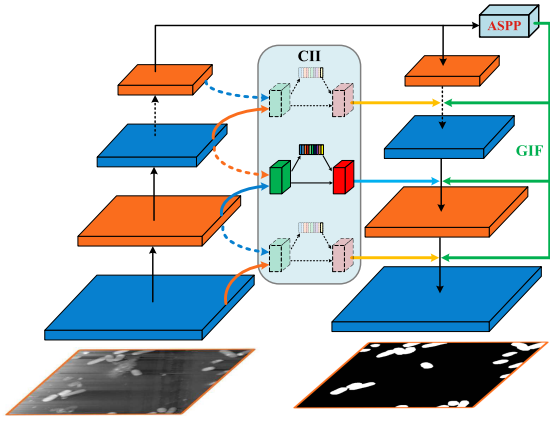


Fig. 3. The overall network structure of the proposed method, including a basic U-shaped backbone, the CII strategy in the middle of the two pathways and the global information flows (GIF) in the right of the top-down pathway. The adopted GIF includes an atrous spatial pyramid pooling (ASPP) module and several global guidance flows (as illustrated by the green lines).

neural network. Therefore, the U-shaped structure is adopted as the backbone of the proposed segmentation algorithm.

A. Centralized Information Interaction Strategy

Although the U-shaped neural network has the above advantages, it still needs to be further improved. Specifically, after an image is down-sampled by convolution and pooling operations and then up-sampled to the same resolution, it inevitably leads to image blur and loss of detailed information. Traditional U-shaped neural network cascades the feature maps of the same resolution in the bottom-up pathway and the top-down pathway as the input of the next layer by skip connections, which reduces the loss of local features to a certain extent. Nevertheless, the skip connections can only obtain the feature information of the corresponding level. For a higher network level in a U-shaped structure, it is easier to acquire richer high-level (global) semantic information and pinpoint the specific location of the target, however, the underlying details are missing at the same time. Based on the above analysis, we consider introducing the centralized information interaction (CII) strategy between the bottom-up pathway and the top-down pathway of the U-shaped network, as shown in the middle of Fig. 3, so as to realize efficient cross-scale information interaction of different scales.

CII is a recently proposed information sharing strategy that focuses on improving the connection between the two pathways of the U-shaped network, and has achieved excellent results in the fields of salient object detection and image segmentation [47]. CII uses a solo module to realize multi-scale information fusion between different feature levels to obtain richer feature information, so that the model can accurately locate high-level positioning information and pay attention to low-level detailed information, thus effectively solving the above-mentioned problems in the U-shaped network. The CII module moves to different places when processing different intermediate features, which is similar to a two-dimensional kernel performing convolution operations on the input image. As shown in Fig. 3, the CII module with solid edges and brighter colors corresponds to the current position of the

module, while the module with dashed edges and darker colors represents other possible positions that the CII module may move to. The specific formula of the CII strategy is as follows:

$$T_j = f_{CII}(F_m, \dots, F_n), 1 \leq j \leq 5, \quad (1)$$

where $f_{CII}(\cdot)$ indicates the interaction function of the identical interactor, which is designed to collaborate with the implementation of the CII strategy. F_m, F_n are the output feature maps of the corresponding level in the bottom-up pathway, which satisfy the constraint: $1 \leq m \leq n \leq 5$. T_j represents the CII output feature map processed by the information interaction module. It is worth noting that, for the sake of simplicity, our network structure only presents four layers, as shown in Fig. 3. When $j = 5, m = n = 5$, the input features of the highest level of the top-down pathway are only provided by the highest level of the bottom-up pathway.

Since the input of the CII strategy has multi-scale features, an attention mechanism is considered to be introduced to prune redundant information. The attention mechanism tends to focus on a part of the image information that assists in judgment, while suppressing other useless information, which greatly facilitates the efficiency of image processing. Therefore, a two-path attention module is designed to cooperate with the specific implementation of the CII strategy (as shown in Fig. 4). The parameters are shared for information interactors dealing with different scales, which can perceive multi-scale information and reduce the number of parameters in the model. The designed two-path attention module consists of a channel attention module (CAM), a spatial attention module (SAM) [54] and several BasicBlocks [55]. The channel attention aims to improve the performance of three aspects: 1) explicitly model the correlation between different channels, 2) automatically obtain the importance of each feature channel through network learning, 3) assign different weight coefficients to each channel to strengthen important features while suppressing unimportant features. The spatial attention aims to improve the feature expression of key regions. Essentially, the spatial information in the original image is transformed into another space and the key information can be retained through the spatial transformation module. Afterwards, a weight mask is generated for each location and the outputs are weighted such that regions of interest are enhanced while irrelevant background regions are weakened. The specific mathematical expression of the designed two-path attention module is as follows:

$$F' = M_c(B(F_{i+1})) \otimes B(F_i), \quad T_j = B(M_s(F') \otimes F'), \quad (2)$$

where $M_c(\cdot)$ and $M_s(\cdot)$ are the weights of channel attention module and spatial attention module, respectively. $\otimes$ represents the element-wise multiplication and F' is an intermediate variable. $B(\cdot)$ is the function of BasicBlock which contains a sequence of two 3×3 convolution layers as shown in the upper right corner of Fig. 4. It might be noted that, the principle for selecting the size of the convolutional kernel is to reduce computational complexity while maintaining the receptive field size with good feature extraction capability. Therefore, the size of the convolutional kernel in BasicBlock is set to 3×3 since it is the smallest size capable of capturing pixel

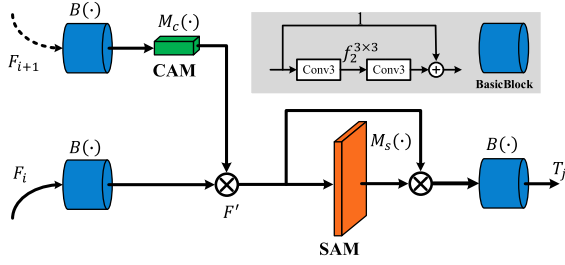


Fig. 4. The schematic diagram of the proposed two-path attention module. The module is made up of a channel attention module, a spatial attention module and three BasicBlocks.

eight neighborhood information. In addition, the large-size convolutional layer can be replaced by the stacking of multiple 3×3 convolutional layers with fewer parameters, which reduces computational and storage overhead while aiding in mitigating overfitting issues. Moreover, smaller convolution kernels can efficiently capture spatial information and smaller details/features in the input.

B. Global Information Flows

On the other hand, as mentioned in the previous analysis, high-level semantic information can be used to locate the specific location of the target area. However, the high-level semantic information is gradually diluted due to the up-sampling operation during the top-down transmission process, which affects the final segmentation performance. Therefore, the GIF is adopted to guide the high-level semantic information to each level of the top-down pathway [48], so as to ensure that the high-level semantic information is not diluted and rich low-level detailed information is retained. The adopted GIF is shown in the right part of Fig. 3, which includes an ASPP module and several global guidance flows. The specific formula is as follows:

$$T'_j = f_{up}(ASPP(F_5)), 1 \leq j \leq 5, \quad (3)$$

where $f_{up}(\cdot)$ indicates the up-sampling operation. F_5 represents the semantic information extracted from the highest level of the bottom-up pathway.

ASPP is a network module originally from the DeepLabv2 network [56], and is later improved and enhanced in the DeepLabv3 network [57]. The function of this module is to increase the receptive field of the network without down-sampling, and enhance the ability of the network to capture multi-scale context information. ASPP designs several dilated convolutions with different sampling rates to capture multi-scale information, which can increase the receptive field without losing resolution and improve segmentation accuracy. T'_j refers to the output result after up-sampling operation on the information extracted by ASPP at a certain magnification. As a global information extractor in the proposed GIF, the multi-scale information extracted by ASPP is sent to each level of the top-down pathway through a series of global guidance flows, which solves the problem that the high-level semantic information is gradually diluted during the top-down transmission process. GIF ensures that the output layer of the network has enough deep semantic information to accurately

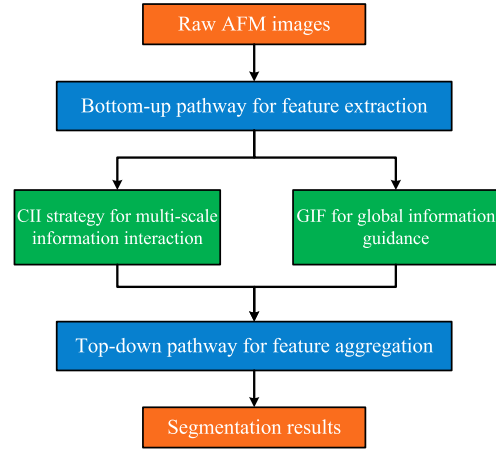


Fig. 5. The general flow chart of the proposed segmentation algorithm.

locate the specific location of the segmented object, which improves the segmentation accuracy of the network.

To better illustrate the inner logic of the proposed algorithm, the general flow chart of the algorithm is drawn as Fig. 5. For a raw AFM image, it is first down-sampled by the bottom-up pathway for feature extraction, which is gradually up-sampled to the original size of the input by the top-down pathway. Furthermore, the CII strategy in the middle of the two pathways is adopted to achieve multi-scale information fusion between different feature levels, so that the model can accurately locate high-level positioning information and pay attention to low-level detailed information. On the other hand, since the high-level semantic information is gradually diluted due to the up-sampling operation during the top-down transmission process, the GIF is utilized to guide the high-level semantic information to each level of the top-down pathway, thus ensuring that the high-level semantic information is not diluted and the rich low-level detailed information is retained. Afterwards, the feature aggregation is implemented by the top-down pathway, based on which accurate AFM image segmentation is achieved by the proposed algorithm.

IV. EXPERIMENT AND APPLICATION

A. Dataset Construction

Methods based on deep learning often require large amounts of data to fully train the model. However, the AFM research lacks enough publicly available datasets for model training, which becomes a big obstacle to develop deep-learning-related algorithms. Therefore, a dataset containing 500 images and the corresponding label masks is constructed to train the proposed neural network, which could also facilitate further progress of deep-learning-based AFM image processing algorithms, such as deep-learning-based classification and reconstruction tasks. The built AFM image dataset is mainly composed of two parts, 200 images scanned by our own and 300 images collected from SSNOMBACTER [22]. The SSNOMBACTER dataset is made up of fifteen kinds of bacterial species with different scanning ranges. These images are denoised by the contributors via using digital restoration methods, artifacts, distortion and noise are thus effectively removed as shown in

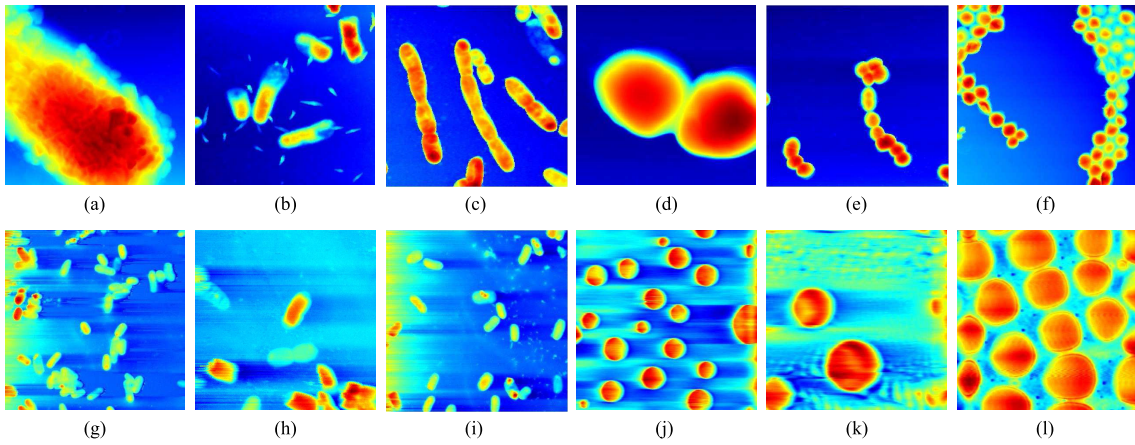


Fig. 6. An overall exhibition of the dataset. (a)–(f) are six representative images collected from the SSNOMBACTER dataset, which include round and rod-shaped bacterial cells of different sizes, the imaging ranges of (a) and (d) are $2 \mu\text{m} \times 2 \mu\text{m}$ while the ranges of (b), (c), (e) and (f) are $10 \mu\text{m} \times 10 \mu\text{m}$. (g)–(l) show six representative self-scanned AFM images, where (g)–(i) are the images of *Escherichia coli* cells of different scanning ranges and resolution, (j)–(l) correspond to PS-LDPE samples scanned with different frequencies, ranges and resolution. The adopted PS-LDPE sample is a mixture of Polyolefin Elastomer and Polystyrene coated onto a silicon substrate. For the self-scanned AFM images, the scanning ranges of (h) and (k) are $5 \mu\text{m} \times 5 \mu\text{m}$, the scanning ranges of others are $10 \mu\text{m} \times 10 \mu\text{m}$.

Figs. 6(a)–(f). These images have flat substrates and relatively clear boundaries. The remaining 200 images are scanned by a Nanosurf Flex-FPM AFM system using constant-force contact mode with vertical feedback. *Escherichia coli* samples and PS-LDPE samples are scanned with different ranges, frequencies and resolution. These self-scanned raw images contain multiple artifacts and distortion as illustrated in Figs. 6(g)–(l), and they are then normalized so that the minimum grayscale value is 0 and the maximum grayscale value is 255. Afterwards, the Colabeler software is utilized to manually segment each image as labels. Moreover, this dataset is randomly divided into a training set with 375 images and a testing set with 125 images.

B. Loss Function and Evaluation Criteria

Binary cross entropy (BCE) loss is widely used as the loss function in image segmentation and binary classification tasks because of the robustness. It calculates pixel-level loss in an image, which can be expressed as:

$$L_{BCE}(x, y) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(x_i) + (1 - y_i) \log(1 - x_i)], \quad (4)$$

where x and y are the segmentation result and the ground truth label, separately. N denotes the total number of pixels and i is the index of each pixel.

Intersection over union (IoU) loss has the advantage of scale invariance and can better reflect the degree of overlap. Different from BCE loss which pays more attention to the pixel-level differences, the IoU loss focuses on the similarity of the whole image. The mathematical expression of IoU loss can be described as:

$$L_{IoU}(x, y) = 1 - \frac{\sum_{i=1}^N (y_i \cdot x_i)}{\sum_{i=1}^N (y_i + x_i - y_i \cdot x_i)}. \quad (5)$$

Based on the above analysis, a combination of BCE loss and IoU loss is used as the loss function to focus on the overall

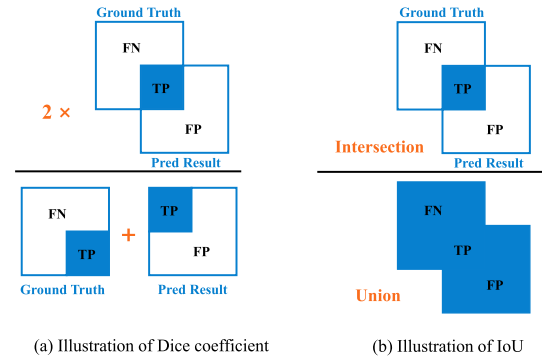


Fig. 7. Illustration of Dice coefficient and IoU.

and partial differences of an image, which can be expressed as follows:

$$L = L_{BCE} + L_{IoU}. \quad (6)$$

Besides, three broadly-used quantitative evaluation indicators in computer vision are utilized to evaluate the performance of the proposed method: namely Dice coefficient, IoU and mean absolute error (MAE). The mathematical expressions of Dice and IoU are as follows:

$$\text{Dice} = \frac{2TP}{(TP + FP) + (TP + FN)}, \quad (7)$$

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \quad (8)$$

the intuitive description of the two evaluation criteria is shown in Fig. 7, where TP is the overlap of the segmentation result with the label image. FN represents the portion of the target region that is not included in the segmented result. FP shows the portion of the non-target region in the segmented area. Both evaluation indicators range from 0 to 1, it implies that the segmentation effect is better when the indicator is closer to 1.

The MAE score calculates the mean pixel-level absolute error between the normalized segmentation result P and the

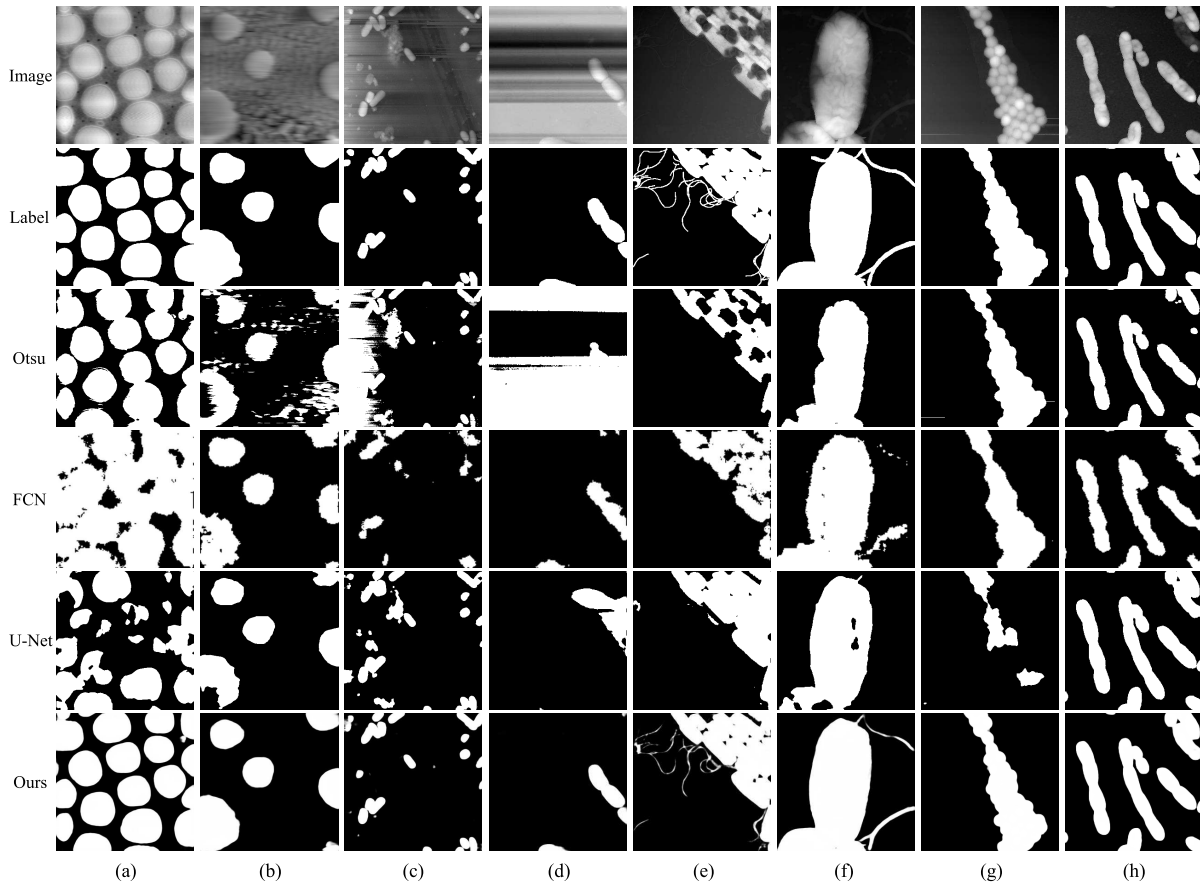


Fig. 8. Qualitative comparison of the segmentation results of different methods. Compared with other methods, the proposed method can accurately segment the objects from the background and preserve more details of the sample.

label image L :

$$MAE = \frac{1}{W \times H} \sum_{x=1}^W \sum_{y=1}^H |P(x, y) - L(x, y)|, \quad (9)$$

where $P(x, y)$ and $L(x, y)$ indicate the corresponding values of pixel (i, j) , W and H are the width and height of the image, respectively. A smaller value of MAE (closer to 0) indicates better performance of the segmentation task.

The proposed deep-learning-based neural network is implemented using the PyTorch library, and an RTX-3070 GPU is utilized for acceleration. The model is trained for 100 epochs with the batch size of 5. The stochastic gradient descent optimizer is adopted for optimization with a learning rate of 0.04, a momentum of 0.9 and a weight decay of 5×10^{-4} . Random cropping is used for data augmentation and the input images are resized to 352×352 in both training and testing processes. It should be noted that no image pre-processing or post-processing is conducted for the proposed method during the whole process.

C. Comparative Experiments Between the Current Methods Without Image Processing and the Proposed Method

We test 125 AFM images in the testing set and achieve good segmentation performance on almost every sample. Some representative results are shown in Fig. 8 to visually compare the proposed method with the contrast methods. It can be

directly observed that the proposed method outperforms the other methods on every sample, all the objects are almost precisely segmented.

The specific analysis of the testing results is presented as follows:

1) The first row in Fig. 8 shows the raw images in the testing set, where the first four images correspond to the self-scanned results with unsatisfactory imaging quality. Specifically, the first image is a relatively desirable PS-LDPE image with multiple specimens. The second image presents a PS-LDPE sample with a smaller scanning range and fewer specimens with lower contrast. The third image is a polluted *Escherichia coli* sample caused by severe vertical drift and illusory slope, resulting in random strips and uneven brightness distribution of the image. The fourth image shows a severely polluted *Escherichia coli* image with a smaller scanning range. The last four images are collected from the SSNOMBACTER dataset with relatively good imaging quality.

2) The images of the second row in Fig. 8 are the corresponding labels, which present the expected segmentation results.

3) The images of the third row in Fig. 8 show the segmentation results of the traditional Otsu method. Since the Otsu method is sensitive to noise and uneven distribution of brightness, the segmentation results are inaccurate, especially for the second to fourth images.

TABLE I

QUANTITATIVE EVALUATION OF DICE, IOU AND MAE OF THE PROPOSED METHOD WITH THE CONTRAST METHODS. MEAN 1 IS THE MEAN INDICATORS OF THE EIGHT SAMPLES IN FIG. 8. MEAN 2 INDICATES THE MEAN INDICATORS OF 125 AFM IMAGES IN THE TESTING SET

Method Sample	Otsu			FCN			U-Net			Ours		
	Dice	IoU	MAE	Dice	IoU	MAE	Dice	IoU	MAE	Dice	IoU	MAE
Fig. 8(a)	0.915	0.844	0.103	0.819	0.694	0.235	0.725	0.569	0.286	0.951	0.906	0.040
Fig. 8(b)	0.687	0.523	0.161	0.927	0.863	0.037	0.879	0.784	0.053	0.950	0.904	0.024
Fig. 8(c)	0.410	0.258	0.226	0.676	0.511	0.063	0.774	0.632	0.050	0.844	0.730	0.016
Fig. 8(d)	0.136	0.073	0.625	0.846	0.734	0.015	0.004	0.002	0.129	0.886	0.796	0.006
Fig. 8(e)	0.743	0.591	0.128	0.864	0.760	0.090	0.914	0.842	0.051	0.902	0.821	0.045
Fig. 8(f)	0.845	0.731	0.118	0.925	0.861	0.061	0.905	0.827	0.076	0.976	0.954	0.028
Fig. 8(g)	0.945	0.895	0.023	0.968	0.939	0.016	0.547	0.377	0.136	0.977	0.956	0.007
Fig. 8(h)	0.894	0.808	0.055	0.934	0.876	0.043	0.974	0.950	0.015	0.963	0.928	0.015
Mean1	0.697	0.590	0.180	0.870	0.780	0.070	0.715	0.623	0.010	0.931	0.874	0.023
Mean2	0.831	0.728	0.082	0.898	0.824	0.043	0.890	0.831	0.049	0.950	0.906	0.016

4) The images in the fourth and fifth rows present the segmentation results of FCN and U-Net. It should be noted that random cropping is also applied in training the FCN and U-Net models for fair comparison, so as to achieve data augmentation. Besides, the FCN and U-Net models are also trained for 100 epochs with the batch size of 5, which is the same as that of the proposed method. It can be seen from the results that, benefiting from the feature extraction of networks, the segmentation results of the second to fourth images with noise are better than the Otsu algorithm. However, caused by limited training data, the segmentation effect for other images is not improved and even worse compared with the Otsu algorithm.

5) The segmentation result of the proposed method is shown in the last row of Fig. 8. All the objects in each image are almost precisely segmented from the background, demonstrating the effectiveness of the proposed method. On the other hand, it also indicates that the proposed method has more meticulous segmentation performance than the contrast methods, which is reflected in the segmented mycelium in the fifth and sixth images. The qualitative comparison indicates that, due to the powerful multi-scale feature interaction ability of the CII strategy and the efficient object localization capacity of GIF, the proposed method can accurately segment the objects from the substrate in different circumstances with strong robustness.

To quantitatively evaluate the validity of the proposed method, Dice coefficient, IoU and MAE of all the segmentation results in Fig. 8 are calculated as shown in TABLE I. The evaluation results also indicate that the proposed method achieves the best performance in most of the cases and achieves rather considerable segmentation results on the rest. Besides, for each method, the mean segmentation indicators of the eight samples in Fig. 8 are calculated and denoted as Mean 1 in TABLE I, the mean indicators of the segmentation results for the whole 125 testing AFM images are also provided as Mean 2 in TABLE I, which demonstrate the robustness and effectiveness of the proposed method. On the other side, the universality of the proposed method makes it high-efficiency

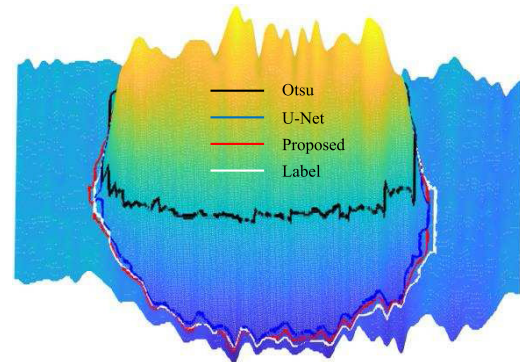


Fig. 9. Mesh plot of the specimen with detected boundaries of different methods. It is clear that the detected boundary of the proposed method is closer to the label boundary.

in segmentation tasks, since it is able to achieve automatic image segmentation without discrimination in approximately 0.05 seconds for each image in the dataset through extensive testing. As a consequence, the proposed approach can run at an average speed of 20 fps to achieve image segmentation in high precision, making it more efficient than traditional methods since no additional image processing is needed for the proposed method.

Moreover, the mesh plot for a selected specimen with the detected boundaries obtained by the proposed method and contrast methods are presented in Fig. 9. The boundaries detected by the Otsu method, U-Net, the proposed method and the label boundary are plotted as the black curve, the blue curve, the red curve and the white curve, separately. It can be obviously observed that the detected boundary of the proposed method is closer to the label boundary (ground truth) than other methods, which further verifies the better image segmentation performance of the proposed method.

D. Comparative Experiments Between the Current Methods With Image Processing and the Proposed Method

Furthermore, to demonstrate the accuracy of the proposed method and the contrast methods with image processing,

TABLE II
QUANTITATIVE EVALUATION OF THE SEGMENTATION RESULTS OBTAINED BY THE CONTRAST METHODS WITH IMAGE PROCESSING AND THE PROPOSED METHOD WITHOUT IMAGE PROCESSING

Method Sample	Otsu			FCN			U-Net			Ours		
	Dice	IoU	MAE	Dice	IoU	MAE	Dice	IoU	MAE	Dice	IoU	MAE
Improved (a)	0.917	0.847	0.102	0.873	0.774	0.169	0.750	0.601	0.296	0.951	0.906	0.040
Original (a)	0.915	0.844	0.103	0.819	0.694	0.235	0.725	0.569	0.286			
Improved (b)	0.782	0.642	0.111	0.932	0.872	0.033	0.930	0.870	0.032	0.950	0.904	0.024
Original (b)	0.687	0.523	0.161	0.927	0.863	0.037	0.879	0.784	0.053			
Improved (c)	0.753	0.604	0.054	0.687	0.523	0.066	0.779	0.638	0.049	0.844	0.730	0.016
Original (c)	0.410	0.258	0.226	0.676	0.511	0.063	0.774	0.632	0.050			
Improved (d)	0.823	0.700	0.017	0.873	0.775	0.015	0.004	0.002	0.129	0.886	0.796	0.006
Original (d)	0.136	0.073	0.625	0.846	0.734	0.015	0.004	0.002	0.129			
Improved (e)	0.845	0.731	0.094	0.878	0.783	0.072	0.914	0.842	0.051	0.902	0.821	0.045
Original (e)	0.743	0.591	0.128	0.864	0.760	0.090	0.914	0.842	0.051			
Improved (f)	0.920	0.852	0.067	0.933	0.875	0.059	0.928	0.866	0.061	0.976	0.954	0.028
Original (f)	0.845	0.731	0.118	0.925	0.861	0.061	0.905	0.827	0.076			
Improved (g)	0.957	0.918	0.018	0.969	0.940	0.014	0.628	0.457	0.126	0.977	0.956	0.007
Original (g)	0.945	0.895	0.023	0.968	0.939	0.016	0.547	0.377	0.136			
Improved (h)	0.929	0.867	0.040	0.936	0.879	0.037	0.974	0.950	0.015	0.963	0.928	0.015
Original (h)	0.894	0.808	0.055	0.934	0.876	0.043	0.974	0.950	0.015			
Improved mean	0.866	0.770	0.081	0.885	0.803	0.058	0.738	0.653	0.095	0.931	0.874	0.023
Original mean	0.697	0.590	0.180	0.870	0.780	0.070	0.715	0.623	0.010			

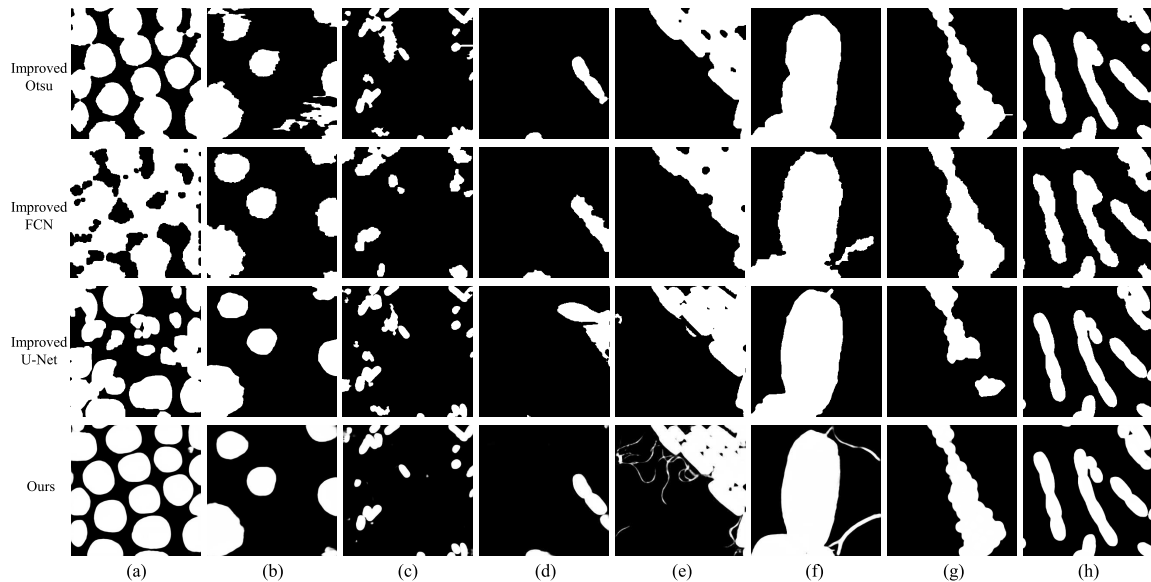


Fig. 10. Comparison of the segmentation results obtained by the contrast methods with image processing and the proposed method without image processing.

another set of comparative experiments are carried out with the results shown in Fig. 10. Specifically, the topography images of the eight samples in the first row in Fig. 8 are used to demonstrate the accuracy of the proposed method over the current methods with image processing. For each contrast method, the suitable image processing technique, including pre-processing or post-processing, is screened out through numerous attempts to obtain the best segmentation results to the maximum extent possible. On this basis, the quantitative accuracy evaluation of different segmentation results is carried out to further demonstrate the effectiveness of the proposed method.

The specific comparison and analysis of different contrast methods are presented as follows:

1) *Otsu method with image processing*: For the segmentation results of the Otsu method without image processing in the third row in Fig. 8, the performance for samples (a), (e)-(h) is relatively good due to the high imaging quality, which can be further improved by image post-processing, including opening operation, closing operation, hole filling and small area removal. For sample (b), image contrast enhancement is first implemented to facilitate further image segmentation. On this basis, the Otsu method combined with image post-processing, *i.e.*, morphological image processing including opening operation, closing operation and hole filling, is used to obtain the improved segmentation result. Furthermore, for sample (c) and sample (d), the distortion of the raw images caused by vertical drift and illusory slope is first corrected [51],

based on which the refined segmentation results are achieved using the Otsu algorithm with image post-processing. Specifically, morphological opening operation, closing operation and small area removal are implemented for the corrected image of sample (c), morphological opening and closing operations are carried out for the corrected image of sample (d). The improved segmentation results obtained by the Otsu method with image processing are shown in the first row in Fig. 10.

2) *FCN with image processing*: The performance improvement of FCN through image pre-processing is first explored, specifically, the enhanced/corrected images are used to train a new FCN model, thus generating the updated segmentation results. However, the model output tends to indicate negligible performance improvement when dealing with pre-processed images, since the segmentation results are almost identical to the images in the fourth row in shown in Fig. 8. Therefore, image post-processing is carried out to refine the segmentation output of FCN, and the polished results via morphological image processing are shown in the second row in Fig. 10.

3) *U-Net with image processing*: Similar to FCN, the output of U-Net model demonstrates the limited improvement in handling with the pre-processed images. Consequently, image post-processing techniques, specifically morphological image processing, are implemented to refine the output of the U-Net model with the segmentation results shown in the third row in Fig. 10.

It can be observed from Fig. 10 that, although image processing is added to the comparison methods to improve segmentation performance, the proposed approach still exhibits better performance as shown in the fourth row in Fig. 10, which is reflected in the segmentation results of the proposed method having more accurate feature regions and richer detailed features.

To quantitatively compare the accuracy of the contrast methods with image processing and the proposed method, Dice coefficient, IoU and MAE of the different images are calculated as shown in TABLE II, where the bold numbers indicate the best performance of different methods for each sample. For each contrast method, the quantitative indicators obtained without/using image processing are presented to verify the performance improvement of image processing on the contrast methods. The quantitative evaluation results also demonstrate that the proposed method still achieves the best performance in most of the cases and relatively good performance on the rest individual samples, thereby further verifying the good performance of the proposed method.

E. Application in Morphology Analysis

To demonstrate the effectiveness and application of the proposed automated AFM image segmentation method in the morphology analysis workflow, two samples from the testing set are selected to implement automated measurements of the morphological parameters. Fig. 11(a) presents the self-scanned image of the PS-LDPE sample with the range of $5 \mu\text{m} \times 5 \mu\text{m}$, Fig. 11(b) shows the image of the *Burkholderia cenocepacia* sample from SSNOMBACTER with the range of $10 \mu\text{m} \times 10 \mu\text{m}$. Afterwards, the specimens in the two images

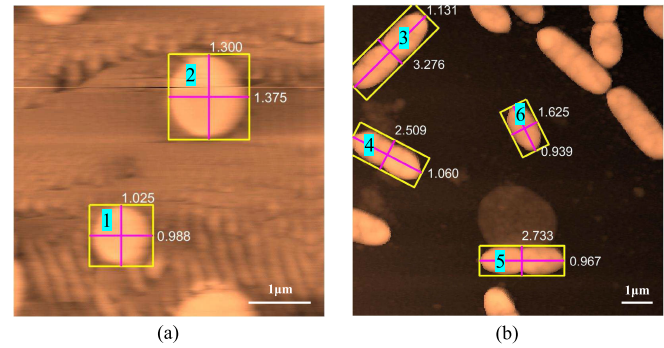


Fig. 11. Morphological measurements of the PS-LDPE sample and the *Burkholderia cenocepacia* sample.

TABLE III
DETAILED PARAMETERS AND CORRESPONDING ERRORS
OF THE SIX MARKED SPECIMENS

Specimen		1	2	3	4	5	6
Length	Measurement/ μm	1.025	1.300	3.276	2.509	2.733	1.625
	Ground truth/ μm	1.062	1.302	3.253	2.500	2.733	1.648
	Relative error	3.48%	0.15%	0.71%	0.36%	0.00%	1.40%
Width	Measurement/ μm	0.988	1.375	1.131	1.060	0.967	0.939
	Ground truth/ μm	0.963	1.396	1.131	1.072	1.000	0.945
	Relative error	2.60%	1.50%	0.00%	1.12%	3.30%	0.63%
Perimeter	Measurement/ μm	3.169	4.222	7.429	5.921	6.327	3.977
	Ground truth/ μm	3.209	4.257	7.432	5.965	6.489	4.173
	Relative error	1.24%	0.83%	0.04%	0.74%	2.50%	4.69%
Area	Measurement/ μm^2	0.808	1.427	3.030	2.144	2.220	1.180
	Ground truth/ μm^2	0.814	1.432	3.011	2.191	2.316	1.214
	Relative error	0.74%	0.35%	0.63%	2.15%	4.15%	2.80%
Aspect ratio	Measurement	1.035	1.058	2.897	2.367	2.826	1.731
	Ground truth	1.028	1.072	2.876	2.332	2.733	1.744
	Relative error	0.68%	1.31%	0.73%	1.50%	3.40%	0.75%
Solidity	Measurement	0.988	0.990	0.961	0.958	0.959	0.982
	Ground truth	0.986	0.988	0.961	0.958	0.966	0.957
	Relative error	0.00%	0.20%	0.00%	0.00%	0.72%	2.61%

are automatically segmented by the proposed method, among which six typical specimens are marked as 1 to 6.

The morphological parameters, including length, width, perimeter, area, aspect ratio and solidity, are automatically calculated based on the segmentation results of the proposed method, which are shown in TABLE III. Specifically, based on the segmentation results, the smallest bounding box of each specimen is generated by the *minboundrect* function in MATLAB. Therefore, the number of pixels occupied by the length and width of the smallest bounding box is determined for each specimen. The length/width of each specimen is calculated by multiplying the pixel number of the length/width by the pixel size, where the pixel size denotes the actual size of a pixel calculated by dividing the scanning range by the resolution in x-axis or y-axis. Similarly, the perimeter, area, aspect ratio and solidity of the specimens are calculated by the *regionprops* library function in MATLAB based on the segmentation results. Moreover, the true morphological parameters are also calculated as the ground truth in TABLE III, thus obtaining the relative errors between the measurements and the true values. To obtain the true parameters, the sample image is manually segmented in pixel level by the Colabeler

software, based on which the true parameters are calculated as the ground truth in the same way as the measurements obtained above. It can be seen from TABLE III that, the measurement errors are all within 5%, many errors are even less than 1%, which further demonstrate the good performance of the proposed morphology analysis strategy.

V. CONCLUSION

This paper proposes an automated segmentation workflow based on an improved deep learning strategy to achieve accurate morphology analysis for AFM images. To achieve cross-scale information interaction between different layers in the traditional U-shaped neural network, the CII strategy is introduced and a two-path attention module is designed as a cooperative module to further inspire the potential of CII. Furthermore, the GIF is adopted to ensure that the global semantic information extracted by the ASPP module is not diluted when transmitted in the top-down pathway. Comparative experiments of different methods demonstrate the higher accuracy and strong robustness of the proposed method. Finally, automated morphological measurements of two samples are conducted to illustrate the validity and application of the proposed analysis workflow. Future work will focus on applying the proposed strategy to more kinds of scenarios, such as specimen number counting and morphological characterization.

ACKNOWLEDGMENT

The authors would like to thank Prof. Xizeng Feng from the College of Life Sciences, Nankai University, for providing *Escherichia coli* cells in this work. They would also like to thank the providers of the SSNOMBACTER dataset.

REFERENCES

- [1] C. Dai, G. Shan, X. Liu, C. Ru, L. Xin, and Y. Sun, "Automated orientation control of motile deformable cells," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 3, pp. 2126–2134, Jul. 2023.
- [2] G. Shan et al., "3D morphology measurement for blastocyst evaluation from 'all angles,'" *IEEE Trans. Biomed. Eng.*, vol. 70, no. 6, pp. 1921–1930, Jun. 2023.
- [3] S. Yuan, Z. Wang, L. Liu, N. Xi, and Y. Wang, "Stochastic approach for feature-based tip localization and planning in nanomanipulations," *IEEE Trans. Autom. Sci. Eng.*, vol. 14, no. 4, pp. 1643–1654, Oct. 2017.
- [4] J. Ren and Q. Zou, "Adaptive-scanning, near-minimum-deformation atomic force microscope imaging of soft sample in liquid: Live mammalian cell example," *Ultramicroscopy*, vol. 186, pp. 150–157, Mar. 2018.
- [5] E. Meijering, A. E. Carpenter, H. Peng, F. A. Hamprecht, and J.-C. Olivo-Marin, "Imagining the future of bioimage analysis," *Nature Biotechnol.*, vol. 34, no. 12, pp. 1250–1255, Dec. 2016.
- [6] L. Liu, Y. Liu, T. Li, L. Li, X. Qian, and Z. Liu, "A feasible method for independently evaluating the mechanical properties of glial LC and RGC axons by combining atomic force microscopy measurement with image segmentation," *J. Mech. Behav. Biomed. Mater.*, vol. 126, Feb. 2022, Art. no. 105041.
- [7] L. Li, W.-W. Huang, X. Wang, Y.-L. Chen, and L. Zhu, "Periodic-disturbance observer using spectrum-selection filtering scheme for cross-coupling suppression in atomic force microscopy," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 3, pp. 2037–2048, Jul. 2023.
- [8] X. Meng, H. Zhang, J. Song, X. Fan, L. Sun, and H. Xie, "Broad modulus range nanomechanical mapping by magnetic-drive soft probes," *Nature Commun.*, vol. 8, no. 1, p. 1944, Dec. 2017.
- [9] T. Hackl, M. Poik, and G. Schitter, "Heterodyne AC Kelvin probe force microscopy for nanoscale surface potential imaging in liquids," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–8, 2023.
- [10] J. D. Adams, B. W. Erickson, J. Grossenbacher, J. Brugger, A. Nievergelt, and G. E. Fantner, "Harnessing the damping properties of materials for high-speed atomic force microscopy," *Nature Nanotechnol.*, vol. 11, no. 2, pp. 147–151, Feb. 2016.
- [11] W. Shang, H. Ren, M. Zhu, T. Xu, and X. Wu, "Dual rotating microsphere using robotic feedforward compensation control of cooperative flexible micropipettes," *IEEE Trans. Autom. Sci. Eng.*, vol. 17, no. 4, pp. 2004–2013, Oct. 2020.
- [12] L. Li, A. J. Fleming, Y. K. Yong, S. S. Aphale, and L. Zhu, "High performance raster scanning of atomic force microscopy using model-free repetitive control," *Mech. Syst. Signal Process.*, vol. 173, Jul. 2022, Art. no. 109027.
- [13] K. Lim et al., "High-speed AFM reveals molecular dynamics of human influenza A hemagglutinin and its interaction with exosomes," *Nano Lett.*, vol. 20, no. 9, pp. 6320–6328, Sep. 2020.
- [14] M. Li, N. Xi, Y. Wang, and L. Liu, "Advances in atomic force microscopy for single-cell analysis," *Nano Res.*, vol. 12, no. 4, pp. 703–718, Apr. 2019.
- [15] B. Xu, V. Saygin, K. A. Brown, and S. B. Andersson, "High-resolution measurement of atomic force microscope cantilever resonance frequency," *Rev. Sci. Instrum.*, vol. 91, no. 12, Dec. 2020, Art. no. 123705.
- [16] M. G. Ruppert, A. J. Fleming, and Y. K. Yong, "Active atomic force microscope cantilevers with integrated device layer piezoresistive sensors," *Sens. Actuators A, Phys.*, vol. 319, Mar. 2021, Art. no. 112519.
- [17] A. Stylianou, M. Lekka, and T. Stylianopoulos, "AFM assessing of nanomechanical fingerprints for cancer early diagnosis and classification: From single cell to tissue level," *Nanoscale*, vol. 10, no. 45, pp. 20930–20945, 2018.
- [18] H. Xie and S. Régnier, "High-efficiency automated nanomanipulation with parallel imaging/manipulation force microscopy," *IEEE Trans. Nanotechnol.*, vol. 11, no. 1, pp. 21–33, Jan. 2012.
- [19] A. Bazaei, Y. K. Yong, and S. O. R. Moheimani, "Combining spiral scanning and internal model control for sequential AFM imaging at video rate," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 1, pp. 371–380, Feb. 2017.
- [20] C. D. Onal, O. Ozcan, and M. Sitti, "Automated 2-D nanoparticle manipulation using atomic force microscopy," *IEEE Trans. Nanotechnol.*, vol. 10, no. 3, pp. 472–481, May 2011.
- [21] N. Koder, D. Yamamoto, R. Ishikawa, and T. Ando, "Video imaging of walking myosin v by high-speed atomic force microscopy," *Nature*, vol. 468, no. 7320, pp. 72–76, Nov. 2010.
- [22] M. Lucidi et al., "SSNOMBACTER: A collection of scattering-type scanning near-field optical microscopy and atomic force microscopy images of bacterial cells," *GigaScience*, vol. 9, no. 11, Nov. 2020, Art. no. gaa129.
- [23] Y. Wang, T. Lu, X. Li, and H. Wang, "Automated image segmentation-assisted flattening of atomic force microscopy images," *Beilstein J. Nanotechnol.*, vol. 9, pp. 975–985, Mar. 2018.
- [24] N. P. Barrera, H. Ge, R. M. Henderson, W. J. Fitzgerald, and J. M. Edwardson, "Automated analysis of the architecture of receptors, imaged by atomic force microscopy," *Micron*, vol. 39, no. 2, pp. 101–110, Feb. 2008.
- [25] P. Klapetek, M. Valtr, D. Nečas, O. Salyk, and P. Dzik, "Atomic force microscopy analysis of nanoparticles in non-ideal conditions," *Nanoscale Res. Lett.*, vol. 6, no. 1, pp. 1–9, Dec. 2011.
- [26] T. Raila, T. Penkauskas, F. Ambrulevičius, M. Jankunec, T. Meškauskas, and G. Valinčius, "AI-based atomic force microscopy image analysis allows to predict electrochemical impedance spectra of defects in tethered bilayer membranes," *Sci. Rep.*, vol. 12, no. 1, pp. 1–11, Jan. 2022.
- [27] Y. Han et al., "A novel transfer learning for recognition of overlapping nano object," *Neural Comput. Appl.*, vol. 34, no. 7, pp. 5729–5741, Apr. 2022.
- [28] S. Venkataraman et al., "Automated image analysis of atomic force microscopy images of rotavirus particles," *Ultramicroscopy*, vol. 106, nos. 8–9, pp. 829–837, Jun. 2006.
- [29] D. Ricci and P. C. Braga, "Recognizing and avoiding artifacts in AFM imaging," in *Atomic Force Microscopy*. Cham, Switzerland: Springer, 2004, pp. 25–37.
- [30] Y. Wang, H. Wang, S. Bi, and B. Guo, "Automatic morphological characterization of nanobubbles with a novel image segmentation method and its application in the study of nanobubble coalescence," *Beilstein J. Nanotechnol.*, vol. 6, pp. 952–963, Apr. 2015.

- [31] Y. Wang, T. Lu, X. Li, S. Ren, and S. Bi, "Robust nanobubble and nanodroplet segmentation in atomic force microscope images using the spherical Hough transform," *Beilstein J. Nanotechnol.*, vol. 8, pp. 2572–2582, Dec. 2017.
- [32] B. P. Marsh, N. Chada, R. R. S. Gari, K. P. Sigdel, and G. M. King, "The Hessian blob algorithm: Precise particle detection in atomic force microscopy imagery," *Sci. Rep.*, vol. 8, no. 1, pp. 1–12, Jan. 2018.
- [33] N. Borodinov, W.-Y. Tsai, V. V. Korolkov, N. Balke, S. V. Kalinin, and O. S. Ovchinnikova, "Machine learning-based multidomain processing for texture-based image segmentation and analysis," *Appl. Phys. Lett.*, vol. 116, no. 4, Jan. 2020, Art. no. 044103.
- [34] Y. Kim, J. Lim, and D. Kim, "Accelerating AFM characterization via deep-learning-based image super-resolution," *Small*, vol. 18, no. 3, Jan. 2022, Art. no. 2103779.
- [35] K.-W. Huang, H.-C. Chen, S.-A. Lee, and L.-C. Fu, "Improving 3D recovery based on super-resolution generative adversarial network and uniform continuous trajectory for atomic force microscopy," in *Proc. Amer. Control Conf. (ACC)*, May 2021, pp. 2601–2606.
- [36] O. M. Gordon, J. E. A. Hodgkinson, S. M. Farley, E. L. Hunsicker, and P. J. Moriarty, "Automated searching and identification of self-organized nanostructures," *Nano Lett.*, vol. 20, no. 10, pp. 7688–7693, Oct. 2020.
- [37] S. Prasad et al., "Atomic force microscopy detects the difference in cancer cells of different neoplastic aggressiveness via machine learning," *Adv. NanoBiomed Res.*, vol. 1, no. 8, Aug. 2021, Art. no. 2000116.
- [38] J. Sotres, H. Boyd, and J. F. Gonzalez-Martinez, "Enabling autonomous scanning probe microscopy imaging of single molecules with deep learning," *Nanoscale*, vol. 13, no. 20, pp. 9193–9203, 2021.
- [39] P. Müller, S. Abuhattum, S. Möllmert, E. Ulbricht, A. V. Taubenberger, and J. Guck, "Nanite: Using machine learning to assess the quality of atomic force microscopy-enabled nano-indentation data," *BMC Bioinf.*, vol. 20, no. 1, pp. 1–9, Dec. 2019.
- [40] B. Alldritt et al., "Automated structure discovery in atomic force microscopy," *Sci. Adv.*, vol. 6, no. 9, Feb. 2020, Art. no. eaay6913.
- [41] G. Callari et al., "The influence of uncertainty contributions on deep learning architectures in vision-based evaluation systems," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 7, pp. 2425–2432, Jul. 2019.
- [42] H. Bai and S. Wu, "Nanowire detection in AFM images using deep learning," *Microsc. Microanalysis*, vol. 27, no. 1, pp. 54–64, Feb. 2021.
- [43] S. Farley et al., "Improving the segmentation of scanning probe microscope images using convolutional neural networks," *Mach. Learn., Sci. Technol.*, vol. 2, no. 1, Dec. 2020, Art. no. 015015.
- [44] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.
- [45] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.* Cham, Switzerland: Springer, 2018, pp. 3–11.
- [46] J. Zhang et al., "LCU-Net: A novel low-cost U-Net for environmental microorganism image segmentation," *Pattern Recognit.*, vol. 115, Jul. 2021, Art. no. 107885.
- [47] J.-J. Liu, Z.-A. Liu, P. Peng, and M.-M. Cheng, "Rethinking the U-shape structure for salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 9030–9042, 2021.
- [48] J.-J. Liu, Q. Hou, Z.-A. Liu, and M.-M. Cheng, "PoolNet+: Exploring the potential of pooling for salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 887–904, Jan. 2023.
- [49] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
- [50] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Munich, Germany, Oct. 2015, pp. 234–241.
- [51] Y. Wu, Y. Fang, Z. Fan, and C. Liu, "Accurate morphology characterization using atomic force microscopy via vertical drift correction and illusory slope elimination," *Microsc. Microanalysis*, vol. 27, no. 6, pp. 1366–1374, Dec. 2021.
- [52] A. Belianinov et al., "Big data and deep data in scanning and electron microscopies: Deriving functionality from multidimensional data sets," *Adv. Struct. Chem. Imag.*, vol. 1, no. 1, pp. 1–25, Dec. 2015.
- [53] S. Jesse et al., "Big data analytics for scanning transmission electron microscopy ptychography," *Sci. Rep.*, vol. 6, no. 1, pp. 1–8, May 2016.
- [54] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
- [55] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [56] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
- [57] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.



Yingao Chang received the B.S. degree in automation from Jilin University, Changchun, China, in 2021. He is currently pursuing the M.S. degree in artificial intelligence with the College of Artificial Intelligence, Institute of Robotics and Automatic Information System, Nankai University, Tianjin, China. His research interests include nanomanipulation and imaging of atomic force microscopy.



Zhiang Liu received the B.S. degree from the Harbin Institute of Technology, Harbin, China, in 2019, and the M.S. degree from the College of Artificial Intelligence, Institute of Robotics and Automatic Information System, Nankai University, Tianjin, China, in 2022, where he is currently pursuing the Ph.D. degree in artificial intelligence. His research interests include machine learning and computer vision.



Yinan Wu (Member, IEEE) received the B.S. degree in intelligence science and technology from Nankai University, Tianjin, China, in 2013, and the Ph.D. degree from the Institute of Robotics and Automatic Information System, Nankai University, in 2018.

From 2019 to 2021, he was a Post-Doctoral Fellow with the College of Artificial Intelligence, Institute of Robotics and Automatic Information System, Nankai University, where he is currently an Associate Professor. His research interests include nanopositioning and imaging of the atomic force microscopy.



Yongchun Fang (Senior Member, IEEE) received the B.S. degree in electrical engineering and the M.S. degree in control theory and applications from Zhejiang University, Hangzhou, China, in 1996 and 1999, respectively, and the Ph.D. degree in electrical engineering from Clemson University, Clemson, SC, USA, in 2002.

From 2002 to 2003, he was a Post-Doctoral Fellow with the Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, NY, USA.

He is currently a Professor with the College of Artificial Intelligence, Institute of Robotics and Automatic Information System, Nankai University, Tianjin, China. His research interests include AFM-based nano-systems, visual servoing, and control of under-actuated systems, including overhead cranes.